DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-500-01 — Replace 3 supports on 8"-P150-FL-2009

Area	A500 Pipe rack	**Task**	SP Pipe support replacement
Priority	3 (reliability)	**Issued by**	Facility owner (demonstration)
Primary asset	**rack-bent:720** — Pipe rack bent at E 60'-0" — E 60'-0" N 0.000" EL 122'-0"		
Equipment	—	**Lines**	8"-P150-FL-2009
Isometrics	ISO-8"-P150-FL-2009 (generated); owner's drawings ISO-FL-2009		
Permits	General work permit, Hot work permit		

Reason

Corroded shoes and guides on the flare header.

Scope of work

1. Support 8"-P150-FL-2009 temporarily either side of each support location.
2. Cut out guide 8"-P150-FL-2009/S1 at E 40'-0" N 4'-6" EL 122'-0"; prepare surfaces.
3. Cut out shoe 8"-P150-FL-2009/S2 at E 60'-0" N 4'-6" EL 122'-0"; prepare surfaces.
4. Cut out guide 8"-P150-FL-2009/S3 at E 80'-0" N 4'-6" EL 122'-0"; prepare surfaces.
5. Weld new supports to WS-03; VT and MT of every fillet weld (SW-NDE-3).
6. Remove temporary supports; check the line rests on every support.

Affected items: 8"-P150-FL-2009/S1, 8"-P150-FL-2009/S2, 8"-P150-FL-2009/S3

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-3, SW-PS-12, SW-WO-1, SW-WO-2, SW-WO-7, SW-WPS-1

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-FL-2009	8"-P150-FL-2009	8"-P150-FL-2009	5	0 error(s), 1 warning(s) — see below

ISO-FL-2009 (PDF) — the generated isometrics follow the ITP.

Drawing	Sheet	Level	Finding
ISO-FL-2009	1	warning	weld 8"-P150-FL-2009/W2 (E 21'-11 5/16" N 4'-6" EL 123'-0", sheet 1) has no W2 label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
rack-bent:720	structure	Pipe rack bent at E 60'-0"		E 60'-0" N 0.000" EL 122'-0"
8"-P150-FL-2009/S1	support	GUIDE FOR 8" LINE (GUIDE-8)	8"-P150-FL-2009	E 40'-0" N 4'-6" EL 122'-0"
8"-P150-FL-2009/S2	support	SHOE FOR 8" LINE (SHOE-8)	8"-P150-FL-2009	E 60'-0" N 4'-6" EL 122'-0"
8"-P150-FL-2009/S3	support	GUIDE FOR 8" LINE (GUIDE-8)	8"-P150-FL-2009	E 80'-0" N 4'-6" EL 122'-0"

Parts

Description	Qty	Replaces
GUIDE FOR 8" LINE (GUIDE-8)	1	8"-P150-FL-2009/S1
SHOE FOR 8" LINE (SHOE-8)	1	8"-P150-FL-2009/S2
GUIDE FOR 8" LINE (GUIDE-8)	1	8"-P150-FL-2009/S3

New welds (SW-WO-4)

Weld	Replaces	On drawing	Line	Size	Wall	Shop/field	Spec	NDE	Location	Welder	Date
WO-500-01-W01	8"-P150-FL-2009/S1	—	8"-P150-FL-2009	8" P150	—	field	WS-03	VT + MT	E 40'-0" N 4'-6" EL 122'-0"		
WO-500-01-W02	8"-P150-FL-2009/S2	—	8"-P150-FL-2009	8" P150	—	field	WS-03	VT + MT	E 60'-0" N 4'-6" EL 122'-0"		

Weld	Replaces	On drawing	Line	Size	Wall	Shop/field	Spec	NDE	Location	Welder	Date
WO-500-01-W03	8"-P150-FL-2009/S3	—	8"-P150-FL-2009	8" P150	—	field	WS-03	VT + MT	E 80'-0" N 4'-6" EL 122'-0"		

Welding specification WS-03 — Fillet welds of support attachments

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Scope	Shoes, guides and anchors to pipe or steel
Joint	Fillet, leg per the support part (SW-PS-12)
Process	Shielded metal arc throughout
Filler	SW-F2 covered electrode
Preheat	≥ 30 °F above dew point and ≥ 60 °F
Interpass	≤ 450 °F
Post-weld heat treatment	None
Rules	SW-WPS-1, SW-WPS-2, SW-WPS-3, SW-PS-12

NDE

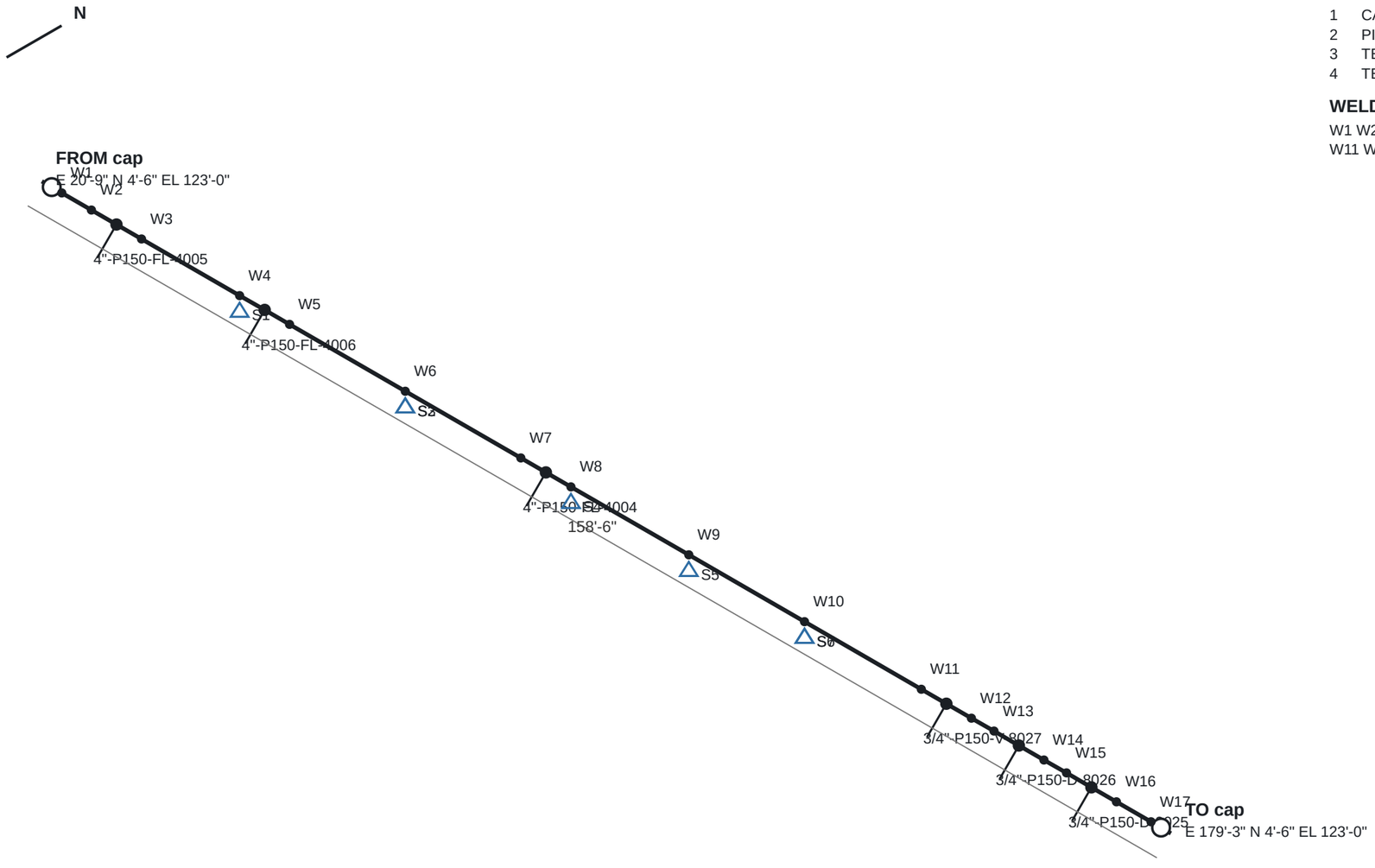
Method	Extent	Welds	Basis
VT + MT	100 % of attachment fillet welds	WO-500-01-W01, WO-500-01-W02, WO-500-01-W03	SW-NDE-3

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Temporary supports in place; line load taken	Owner rigging procedure	Line held at its elevation	Daily report	H	W	—				
2	Old supports removed; pipe surface inspected under the support	SW-NDE-4	No wall loss beyond the owner's criteria (else raise a spool work order)	Inspection record	W	H	—				
3	New supports welded to WS-03	WS-03, SW-WPS-3	Fit and leg as specified	Weld record	S	S	—				
4	VT and MT of every attachment weld	SW-NDE-3	No crack or lack of fusion	VT/MT reports	W	R	R				
5	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 8"-P150-FL-2009 — Replace 3 supports on 8"-P150-FL-2009



BILL OF MATERIALS

1	CAP 8" P150	2
2	PIPE 8" P150 BEVELLED ENDS	151'-4"
3	TEE REDUCING 8" × 4" P150	3
4	TEE REDUCING 8" × 3/4" P150	3

WELDS 17 (0 FIELD) · SUPPORTS 7 · SPOOLS 1

W1 W2 W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13 W14 W15 W16 W17

8"-P150-FL-2009

CLASS F150 — Flare and relief outlets
SERVICE FL · 8" · P150 · DESIGN 150 psi
cap → cap
PIPE 151'-4" · WEIGHT 3684 lb

DWG ISO-8"-P150-FL-2009 REV 0 WO-500-01

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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